

donchian_breakout - Strategy Walkthrough

Strategy name donchian_breakout Files reviewed - src/strategies/donchian_breakout/strategy.p src/strategies/donchian_breakout/backtest_donchian_breakout_v2.py Step-by-step execution flo Backtest fetches OHLCV data, computes Donchian levels, and optional ATR series. 2) Backtester initialized with fixed SL/TP settings and optional ATR multipliers. 3) Loop starts after donchian_lookback + 1 bars. 4) For each bar: bt.on_bar is called first. 5) Signal is computed check_signal on slice_df = df.iloc[:i] (data up to previous bar). 6) check_signal returns LONG breakout above donchian_high and EXIT from LONG on break below donchian_low. 7) If ATR mode i entries are skipped until atr_value is not NaN. 8) Signal is executed with bt.on_signal(..., atr_value=atr_value). 9) Final outputs: stats, equity curve image, report, and trades CSV. P and defaults - Strategy-specific: donchian_lookback (required, no default). - ATR options: atr_period=None, atr_sl_mult=None, atr_tp_mult=None. - Execution and cost: initial_balance=10 commission_pct=0.001, slippage_pct=0.001. - Position controls: position_mode="all_in", trade position_pct=None, pyramiding=1. - Direction and risk: allow_short=False, stop_loss_pct=0.02, take_profit_pct=None. Indicators/calculations - donchian_high = rolling max(high, lookback). donchian_low = rolling min(low, lookback).shift(1). - Optional ATR from src.core.ta.compute_a based stop/take-profit logic in BacktesterV2. Entry conditions - LONG when close > donchian conditions - EXIT long when close < donchian_low. Risk and position management (SL/TP/ATR/trailing/pyramiding/position sizing) - Default stop_loss_pct=0.02 passed to Backte Optional ATR stop/take-profit via atr_sl_mult and atr_tp_mult if atr_period is provided. - No trailing-stop logic in this strategy runner. - Pyramiding supported by BacktesterV2 (default Position sizing handled by BacktesterV2 mode and optional position_pct. Outputs (trades, sta reports) - clean_stats from bt.stats(). - Equity curve PNG under static/backtests/donchian_breakout/<run_id>/. - Trades CSV with run metadata. - QuantStats re enabled. Known ambiguities or inconsistencies in the code - No explicit empty-data guard aft fetch_ohlcv; compute_donchian is called directly. - Signal is computed from df.iloc[:i] (excl current bar), but order executes on current bar close; this introduces a one-bar timing offse with common same-bar-close signal logic.

donchian_breakout - Data note

No trade CSV was produced for this strategy on the synthetic baseline dataset. This usually means the synthetic baseline dataset did not trigger entries under default parameters.